

Matrices : Types, Operation, Properties

* Introduction to Matrices

→ A matrix is a 2d array of numbers arranged in rows or columns.

$$A = \begin{matrix} \text{Rows (m)} \downarrow & \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} \\ & \xrightarrow{\text{Columns (n)}} \end{matrix}$$

→ order of matrix = $m \times n$
= 3×4

→ Given a matrix A of shape (3,4)

→ we can say that

① A is a stack of 4 column vectors

$$\begin{bmatrix} | & | & | & | \\ c_1 & c_2 & c_3 & c_4 \\ | & | & | & | \end{bmatrix}$$

② A is a stack of 3 row vectors

$$\begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ - & r_3 & - \end{bmatrix}$$

→ If $m=n$ then it is square matrix else rectangular matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Square matrix

$$B = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

Rectangular matrix

→ The principal diagonal of a square matrix is the diagonal from the upper left corner to the lower right corner.

→ The secondary diagonal of a square matrix is the diagonal from the lower left corner to the upper right corner.

Principal diagonal

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

secondary diagonal

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

→ Upper Triangular Matrix: A square matrix in which all the element below the principal diagonal is 0.

Upper Triangular Matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

→ Lower Triangular Matrix:

A square matrix in which all the element above the principal diagonal is 0.

Lower Triangular Matrix

$$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Both lower & Upper Triangular Matrix

$$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

→ Diagonal matrix: A square matrix in which all the element other than elements of principal diagonal is 0

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diagonal matrix examples

$$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

→ Identity Matrix: A diagonal matrix with all elements of principal diagonal equal to 1.

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

→ Transpose Matrix: Transpose (A^T) of matrix A is obtained by switching rows as columns and columns as rows.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3}$$

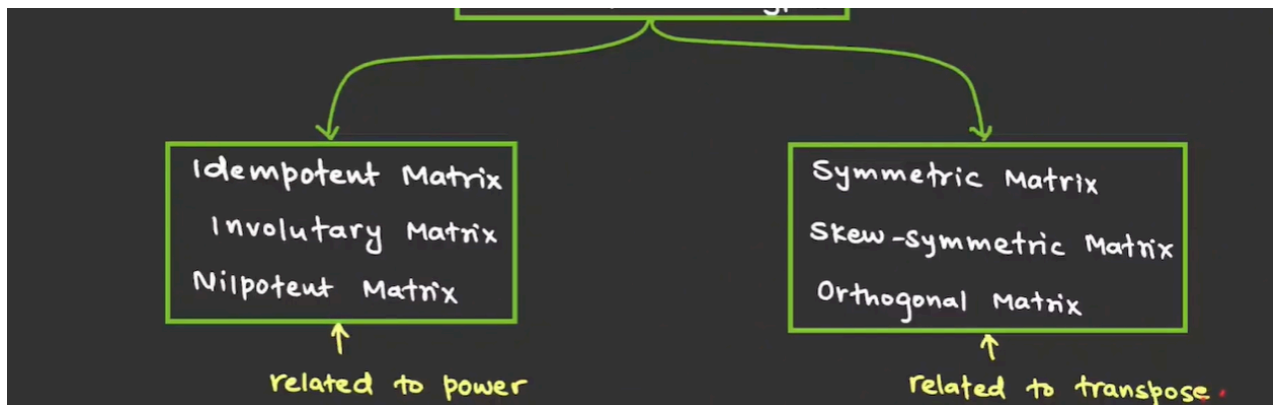
$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}_{3 \times 3}$$

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$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2 \times 3}$$

$$A^T = \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix}_{3 \times 2}$$

Advanced Matrix Types



→ Idempotent Matrix: A square matrix A is idempotent matrix if $A^2 = A$

→ Involutory Matrix: A square matrix A is involutory matrix if $A^2 = I$

→ Nilpotent Matrix: A square matrix A is nilpotent if $A^k = 0$, for some integer k

→ Symmetric Matrix: A square matrix A is symmetric if $A^T = A$

→ Skew-Symmetric Matrix: A square matrix A is skew-symmetric if $A^T = -A$
 → diagonal elements are always 0

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \leftarrow \text{Symmetric Matrix}$$

$$B = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} \leftarrow \text{Skew-Symmetric Matrix}$$

→ Orthogonal Matrix: A square matrix A is orthogonal if $A^T = A^{-1}$

Operation On Matrices :

① Scalar Multiplication: Multiply every element by a scalar k

$$2 \times \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 8 \end{bmatrix}$$

② Matrix Addition: Two matrix can be added only if they have same order

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

③ Matrix Multiplication:

$$A_{m \times n} \times B_{n \times p} = C_{m \times p}$$

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj} = \text{dot product of } i^{\text{th}} \text{ row of } A \text{ and } j^{\text{th}} \text{ column of } B$$

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ Compute AB

$$\rightarrow AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times 7 & 1 \times 6 + 2 \times 8 \\ 3 \times 5 + 4 \times 7 & 3 \times 6 + 4 \times 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

* Key Properties of Matrix Multiplication

- ① $AB \neq BA$ ← not commutative
- ② $(AB)C = A(BC)$ ← Associative
- ③ $A(B+C) = AB + AC$ ← Distributive

④ Trace of Matrix: sum of diagonal elements. defined for a square matrix

$$\text{tr}(A) = \sum a_{ii}$$

⑤ Determinant of Matrix: A scalar value defined for a square matrix

Example: Compute $|A|$ where $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$$\rightarrow |A| = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} - 0 \cdot \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} + 0 \cdot \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 2$$

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⑤ Inverse of a Matrix (A^{-1}): A square matrix A is invertible only if $|A| \neq 0$

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$$

$$\text{Adj}(A) = \text{Cof}(A)^T$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Cofactor ← Minor ←

Example: Compute A^{-1} where $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ ←

$$\rightarrow |A| = 1(2-0) - 0(0-0) + 0(0-0) = 2$$

$$C = \begin{bmatrix} +(1 \times 2) & -(0 \times 0) & +(0 \times 0) \\ -(0 \times 0) & +(1 \times 2) & -(0 \times 0) \\ +(0 \times 0) & -(0 \times 0) & +(2 \times 1) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}^T = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Rank Of A Matrix :

→ Rank of a matrix tells no of Linearly independent rows and columns in a matrix

Example: Compute rank of $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix}$

c₁ c₂ c₃

$$\rightarrow c_3 = 3c_1 + 3c_2$$

Finally, we find that total 2 LI Columns

$$\text{So rank} = 2$$